2c. Content of topics P1 to P9

Topic P1: Matter

P1.1 The particle model

Summary

Knowledge and understanding of the particle nature of matter is fundamental to physics. Learners need to have an appreciation of matter in its different forms, they must also be aware of subatomic particles, their relative charges, masses and positions inside the atom. The structure and nature of atoms are essential to the further understanding of physics. The knowledge of subatomic particles is needed to explain many phenomena, for example the transfer of charges, as well as radioactivity. (Much of this content overlaps with that in the GCSE (9–1) in Chemistry A (Gateway Science content.)

Underlying knowledge and understanding

Learners should be aware of the atomic model, and that atoms are examples of particles. They should also know the difference between atoms, molecules and compounds. Learners should understand how density can be affected by the stat materials are in.

Common misconceptions

Learners commonly confuse the different types of particles (subatomic particles, atoms and molecules) which can be addressed through the teaching of this topic They commonly misunderstand the conversions between different units used in the measurement of volume.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.
All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
PM1.1i	recall and apply: $\text{density (kg/m}^3\text{)} = \dfrac{\text{mass (kg)}}{\text{volume (m}^3\text{)}}$	M1a, M1b, M1c, M3b, M3c, M5c

Topic content		Opportunities to cover:		Practical suggestions
Learning outcomes	To include	Maths	Working scientifically	
P1.1a describe how and why the atomic model has changed over time	the Thomson, Rutherford (alongside Geiger and Marsden) and Bohr models	M5b	WS1.1a, WS1.1c, WS1.1g	Timeline showing the development of atomic theory. Discussion of the different roles played in developing the atomic model and how different scientists worked together.
P1.1b describe the atom as a positively charged nucleus surrounded by negatively charged electrons, with the nuclear radius much smaller than that of the atom and with almost all of the mass in the nucleus		M5b	WS1.1b	Model making (including 3D) of atomic structures.
P1.1c recall the typical size (order of magnitude) of atoms and small molecules	knowledge that it is typically 1×10^{-10}m	M1b	WS1.1d	
P1.1d define density			WS1.2b, WS1.2c, WS1.3c, WS1.3d, WS1.4a, WS1.4b, WS1.4e, WS1.4f, WS2a, WS2b, WS2c, WS2d	Measurement of length, volume and mass and using them to calculate density. (PAG P1) Investigation of Archimedes' Principle using eureka cans. (PAG P1)
P1.1e explain the differences in density between the different states of matter in terms of the arrangements of the atoms and molecules		M5b	WS1.1b	
P1.1f apply the relationship between density, mass and volume to changes where mass is conserved		M1a, M1b, M1c, M3c		